

RESEARCH PROPOSAL

Research title (English)	Evaluating the Efficiency of the Guide-AI-Arch Model in Software Design and Architecture Education via Empirical Technical Metrics
Sub-committee	Information Technology (Information Systems / Software Engineering)
Group name	Nhóm 8
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Abstract	This study proposes the Guide-AI-Arch model, an empirical extension of the foundational Guide-AI-Ed framework (2026), specifically tailored to mitigate Generative AI (GenAI) academic shortcuts in Software Design and Architecture education. The core innovation of this proposal is the transition from subjective, qualitative student self-reports toward the integration of automated hard technical metrics to objectively evaluate students' actual critical verification performance. Utilizing a Randomized Controlled Trial (RCT) research design with 80 computing students, architectural artifacts will be compiled and evaluated using automated static analysis tools including ArchUnit and SonarQube. Structural metrics such as coupling coefficients, structural cohesion (LCOM4), and seeded architectural defect detection rates will be tracked longitudinally. The empirical results aim to mathematically prove that rigorous verification frameworks not only enhance student critical thinking but also directly optimize the technical maintainability, modularity, and structural quality of the resulting software solutions. Key words: GenAI, Software Architecture, Guide-AI-Arch, Technical Metrics, ArchUnit, SonarQube.

1. INTRODUCTION

1.1. Literature review

The explosive adoption of Large Language Models (LLMs) in software engineering education has introduced severe pedagogical vulnerabilities, particularly the phenomenon of "Architectural Hallucinations." When prompted with complex, distributed software designs, GenAI models frequently emit over-engineered software systems, introduce covert cyclic dependencies, or systematically violate core object-oriented principles (such as SOLID). Inexperienced students habitually copy-paste these outputs without executing any formal technical analysis.

The foundational Guide-AI-Ed framework (2026) pioneered a design-oriented conceptual model that mapped course layouts to encourage active checking rather than passive consumption. Grounded in an AI-aware adaptation of the Revised Bloom's Taxonomy, the authors successfully shifted student attention from low-tier cognitive domains (Remembering, Understanding, basic Applying) to high-tier domains (Analyzing, Evaluating, Creating)

through prompt journaling and policy clarity. However, a prominent structural limitation of Guide-AI-Ed was its total dependence on subjective psycho-educational variables, such as self-reported 5-point Likert-scale satisfaction surveys and qualitative interview logs.

In the domain of Software Design and Architecture, system quality cannot be measured via emotional perception or subjective satisfaction. Instead, it demands mathematical validation and static analysis. Recent literature introduces severe concerns regarding architectural decay driven by AI, where generated frameworks exhibit intense structural coupling and poor internal cohesion. This study establishes a novel bridge between pedagogical intervention and hard technical metrics, filling a distinct research gap: the absence of empirical, automated software verification metrics to measure the efficacy of educational AI models.

1.2. The necessity of the research

While industry practitioners leverage tools like GitHub Copilot and ChatGPT to accelerate workflows, uncritical student reliance leads to rapid cognitive decay and severe academic shortcuts. Current research emphasizes a stark dichotomy between professional software engineering productivity and student skill acquisition.

The scientific significance of this study lies in its shift from qualitative claims to mathematical proof. By establishing a rigorous framework that forces students to execute automated verification, we address the core vulnerability of AI-assisted engineering education: the uncritical acceptance of flawed structural architectures. This project provides a scalable template for verifying engineering software artifacts in the GenAI era.

1.3. Feasibility of research

- **Research question:** Does the integration of the Guide-AI-Arch pedagogical intervention model significantly optimize the empirical structural code metrics (coupling, cohesion, and flaw isolation rates) of student-designed software architecture systems compared to unconstrained, direct GenAI usage?
- **Research design:** A rigorous 12-week Randomized Controlled Trial (RCT) where student source code repositories and architectural blueprints are parsed dynamically using automated static analysis engines (ArchUnit and SonarQube) to collect hard quantitative metrics.
- **Sample size and population:** The target population consists of $N = 80$ third-year undergraduate software engineering students. The cohort is randomized into two independent groups of $n = 40$ each: the Control Group (CG) with unconstrained GenAI access, and the Experimental Group (EG) bound to the Guide-AI-Arch verification model.
- **Data sources:** Automated software repositories, ArchUnit structural check reports, SonarQube dashboards, and structured Architecture Trade-off Analysis Matrices (ATAM) submitted by students.
- **Budget and funding:** Conducted within university computing lab infrastructure; no additional funding required.
- **Timeframe:** 12 weeks total. Weeks 1-2: Setup and random grouping; Weeks 3-10: Educational intervention and project development; Weeks 11-12: Automated tool processing and statistical hypothesis analysis.
- **Potential challenges and risks:** Variable student baseline programming proficiency. This is mitigated by collecting a baseline pre-test structural assessment to use as a covariate in the statistical testing.

2. RESEARCH OBJECTIVES

The primary research objectives are:

1. To deploy the Guide-AI-Arch architecture verification framework in an active university course.

2. To mathematically measure and evaluate structural code indicators (Afferent/Efferent coupling, LCOM4 cohesion) under controlled GenAI usage conditions.
3. To evaluate human verification effectiveness through an intentionally seeded defect detection analysis rate.

3. RESEARCH SCOPE

The scope is limited to undergraduate software engineering students designing a distributed microservices e-commerce system framework. The technical boundaries are restricted to static package rules, class design principles, and structural architectural properties testable via ArchUnit and SonarQube.

4. APPROACH AND METHOD

A quantitative empirical methodology is applied. Codebases are subjected to static analysis engines. Statistical hypothesis testing is conducted using two-sample Independent T-tests or non-parametric Mann-Whitney U-tests via R software ($p < 0.05$).

- **Null Hypothesis (H_0):** $\mu_{Experimental} = \mu_{Control}$ (No difference in hard technical metrics).
- **Alternative Hypothesis (H_1):** $\mu_{Experimental} > \mu_{Control}$ (Significant architectural quality improvements).

5. RESEARCH PLAN

No.	Date	Task	Output	Person in charge
1	01/09/2026 – 15/09/2026	Cohort randomization and baseline pre-test setup	Student distribution lists, pre-test code metrics	Trần Quốc Huy
2	16/09/2026 – 15/11/2026	Execution of Guide-AI-Arch intervention vs unconstrained access	ATAM documents, repository commit histories	Dương Minh Kiệt & Nguyễn Phúc Thành
3	16/11/2026 – 30/11/2026	Automated parsing via ArchUnit/ SonarQube and R analysis	Comprehensive structural analytics, statistical report	Lê Đăng Nguyên

6. EXPECTED RESULTS

The study expects to provide a mathematical proof of the framework's efficiency: demonstrating a minimum 40% reduction in cyclic structural dependencies, an optimized LCOM4 score close to 1.0, and a Seeded Defect Detection Rate exceeding 85% in the experimental group, proving that rigorous verification translates into highly industrial-grade software artifacts.

7. REFERENCES

[1] V. Goruslu et al., "Guide-AI-Ed: A Conceptual Framework for Re-designing Higher Education Courses in the Era of Generative AI," **Journal of Higher Education and Technology**, 2026.

[2] IEEE Template Reference for Information Technology Sub-committee compliance.

COMMITMENT ON RESPONSIBLE ARTIFICIAL INTELLIGENCE USAGE IN RESEARCH

We are authors of the research paper titled "Evaluating the Efficiency of the Guide-AI-Arch Model in Software Design and Architecture Education via Empirical Technical Metrics", would like to ensure that all members comply

with the rules relevant to Artificial Intelligence (AI) regulations and ethical guidelines in research practices, which are described in details as follows:

- AI is only used as assistant in correcting grammar, structure and phrasing in academic writing;
- AI usage should NOT be engaged with the creation of content or drafting the research paper, production of fake data or literature synthesis;
- AI usage should be checked and confirmed by all authors. Each author shall bear full responsibility in the accuracy, transparency and originality of their sections, including citations and reference;
- All authors should disclose the use of generative AI and AI-assisted technologies to other co-authors, helping them understand where and how AI was applied in research preparation.

Hereby, we would like to confirm and take full responsibility for the accuracy and transparency of the provided details.

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